



 **Time:** 5 minutes

 **Outcome:** Analyze Profit-Maximizing Output & the $MR = MC$ Rule.

Difficulty: ●●○ Intermediate

Profit-Maximizing Output & the $MR = MC$ Rule



THE CORE IDEA

A firm maximizes profit or minimizes loss by producing the quantity where rising marginal cost equals marginal revenue. The local profit maximum occurs where MC rises through MR. Marginal benefit to the firm exceeds marginal cost. Market price becomes the firm's MR, linking market equilibrium to the firm's marginal output choice.



HOW TO RECOGNIZE IT

- Use the firm's marginal-revenue curve and the rising portion of marginal cost.
- Choose the output where marginal cost equals marginal revenue, provided the firm should operate.
- If marginal revenue exceeds marginal cost, another unit raises profit; if marginal cost exceeds marginal revenue, it lowers profit.
- Do not choose output by minimizing average total cost unless that output also satisfies the marginal rule.

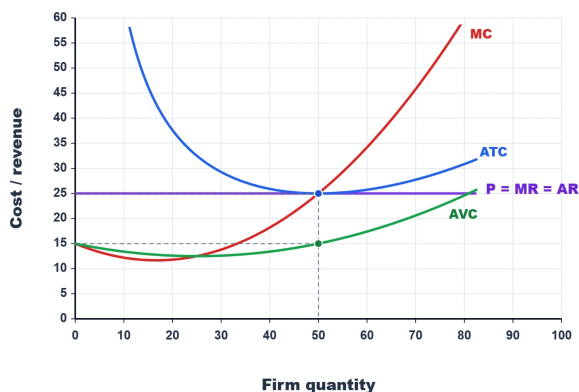


WATCH OUT

A common mistake is to misapply competitive firm rule. Market price becomes the firm's MR, linking market equilibrium to the firm's marginal output choice.



WORKED EXAMPLE: USING THE $MR = MC$ RULE



The firm's market price and marginal revenue are \$25. The rising marginal-cost curve reaches \$25 at 50 units, so the last unit adds exactly as much revenue as cost. Producing fewer units leaves profitable output unmade; producing more makes MC exceed MR. Profit-maximizing output is 50.



CHECK YOURSELF

MR is \$25 and the next unit's MC is \$19. What should the firm do?



READY?

Return to the game and master this concept.